# Muhammad Talha Atif

Full Stack Engineer

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## SUMMARY

Full Stack Engineer with 1+ year of production experience building scalable, fault-tolerant web systems across Python, JavaScript, TypeScript, React.js, and Next.js. Delivered distributed microservices, REST APIs, and full stack features for enterprise and consumer platforms on AWS, with hands-on experience in PostgreSQL, Redis caching, Docker, GitHub Actions CI/CD, and AI-assisted development workflows. Comfortable owning features end to end from database schema to UI, with a consistent focus on testable, maintainable, and secure code.

## WORK EXPERIENCE

**Full Stack Engineer** *Dec '25 – Present*
**MicroTechX** Remote

*Built and maintained full stack features for an enterprise IAM platform, owning backend services, REST APIs, PostgreSQL schema design, and a React/Next.js dashboard across a compliance-heavy, security-first distributed system.*

- Designed fault-tolerant event-driven microservices using Kafka across 8+ distributed services with consumer concurrency tuning, retry windows, and dead-letter queues, ensuring reliable async processing and auditability at scale.
- Built bulk role assignment API processing 5,000+ records per operation using batch queries against PostgreSQL, eliminating synchronous bottlenecks and keeping latency flat regardless of payload size.
- Enforced security best practices across all service endpoints including JWT validation, RBAC, rate limiting, and input sanitisation, and maintained GitHub Actions CI/CD pipelines for automated build, test, and deployment cycles.

**Full Stack Engineer** *Sep '25 – Jan '26*
**Digital Upgraders LLC** Remote

*Shipped full stack features for a production social platform, owning backend APIs, frontend React components, PostgreSQL queries, and AWS-based cloud infrastructure.*

- Resolved a feed API bottleneck by adding selective PostgreSQL indexes, pruning unused DTO fields, optimising queries, and adding a 5-minute in-memory cache, reducing response time from 3 seconds to 700ms across the full stack.
- Built content moderation systems end to end including PostgreSQL schema, REST APIs with ranking logic, and React admin UI, with duplicate-report prevention and edge-case validation enforced at the API layer.
- Led zero-downtime production migration to AWS-backed infrastructure via Fly.io, managing Docker configuration, DNS, Supabase, and environment setup across a live multi-service platform with no user-facing downtime.

## PROJECTS

**Moodly** | AI Experience Booking Platform | Demo

- Built Python FastAPI inference service and React Native frontend end to end, connecting a multimodal emotion detection pipeline to a booking engine via async message queue, with Claude AI and GitHub Copilot used throughout to accelerate development.
- Designed secure, testable REST APIs for the full booking lifecycle with strict input validation, JWT authentication, and unit-tested business logic across all critical paths.

**Memingo** | Language Learning Platform | Demo

- Resolved leaderboard bottleneck by replacing per-request ranking computation with a Redis precomputed cache, reducing PostgreSQL load and cutting API response from 30 seconds to 90ms.
- Built React Native frontend with 9 interactive lesson components and integrated a Gemini AI chatbot with session-aware context management and structured output parsing.

## SKILLS

**Languages** Python, JavaScript, TypeScript, Java, SQL
**Frontend** React.js, Next.js, React Native, Tailwind CSS, Shadcn UI
**Backend** Node.js, Express.js, NestJS, FastAPI, Spring Boot, REST APIs, microservices
**Databases** PostgreSQL, MySQL, MongoDB, Redis, Supabase, Firebase
**Cloud / Infra** AWS (S3, RDS, EC2), Docker, GitHub Actions, CI/CD, Fly.io, Dokploy
**Security** JWT, OAuth 2.0, RBAC, rate limiting, input validation
**AI Tools** Claude AI, GitHub Copilot, Gemini API, OpenAI API

## AWARDS

**Runner-up, App Development Hackathon (2025)** *May '25*
Organized by Punjab University

**Ranked, ICPC Asia Topi Online Programming Contest (2024)** *Nov '24*
Organized by International Collegiate Programming Contests

## EDUCATION

**Bachelor's degree in Computer Science, University of Management and Technology – UMT** *Oct '21 – Aug '25*